Yixuan Shao (邵一轩)

PhD Student · Pulsar/FRB research

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Education

2023–present **PhD student, High-energy astrophysics**, *School of Astronomy and Space Science*, Nanjing University, Advisor: Prof. Ping Zhou.

Working on multiwavelength observations of isolated neutron stars (e.g. magnetars, central compact objects, high-magnetic-field pulsars, radio pulsars), fast radio bursts, and supernova remnants.

2019–2023 **Bachelor of Astronomy**, *School of Astronomy and Space Science*, Nanjing University, Undergrad research advisor: Prof. Ping Zhou.

Study pulsar search with FAST & optical/NIR and X-ray data analysis

Publications

See full list on Ads library

First-author publications

2025 **Shao, Yi-Xuan**, Ping Zhou, Xiao Zhang, Zhi-Yu Zhang, Yang Chen, Qin Han, Di Li, Xiang-Dong Li, Jian-Bin Weng, and Yong Shao. Absence of Radio Emission Reveals an Exceptionally Weak Explosion of the Putative Historical Supernova Pa 30. *ApJ*, volume 992, page L6, October 2025.

2024 **Shao, Yi-Xuan**, Ping Zhou, Xiang-Dong Li, Bin-Bin Zhang, Alberto Javier Castro-Tirado, Pei Wang, Di Li, Zeng-Hua Zhang, Zi-Jian Zhang, You-Dong Hu, and Shashi B. Pandey. GTC Optical/Near-infrared Upper Limits and NICER X-Ray Analysis of SGR J1935+2154 for the Outburst in 2022. *ApJ*, volume 976, page 99, November 2024.

Co-author publications

2026 Shi-Jie Gao, Yi-Xuan Shao, and Xiang-Dong Li. Pulsar Gleaners: Discovery of 19 Pulsars in FAST Archival Data at $|b| < 5°$ and Decl. $< -5°$. *ApJ*, volume 997, page 210, February 2026.

2026 Shi-Jie Gao, Xiang-Dong Li, Song Wang, Kareem El-Badry, De-Jiang Zhou, Yi-Xuan Shao, Zhen Yan, Pei Wang, Ping Zhou, and Jin-Lin Han. Search for Radio Pulsations from Neutron Star Candidates in Detached Binaries. *ApJ*, volume 997, page 222, February 2026.

2025 Shi-Jie Gao, Xiang-Dong Li, Zhen Yan, **Shao, Yi-Xuan**, and Ping Zhou. RRAT J2325–0530: A Rotating Radio Transient with an Atypical Waiting-time Distribution. *ApJ*, volume 991, page 201, October 2025.

2025 Shi-Jie Gao, Xiang-Dong Li, **Shao, Yi-Xuan**, Ping Zhou, Pei Wang, Yun-Wei Yu, Zhen Yan, and Di Li. A Target Search for Fast Radio Bursts Associated with Two Fast Blue Optical Transients: AT2018cow and CSS161010. *ApJ*, volume 990, page 93, September 2025.

2024 Yi-Han Iris Yin, Bin-Bin Zhang, Jun Yang, Hui Sun, Chen Zhang, **Shao, Yi-Xuan**, You-Dong Hu, Zi-Pei Zhu, Dong Xu, Li An, He Gao, Xue-Feng Wu, Bing Zhang, Alberto Javier Castro-Tirado, Shashi B. Pandey, Arne Rau, Weihua Lei, Wei Xie, Giancarlo Ghirlanda, and et al. Piro. Triggering the Untriggered: The First Einstein Probe-detected Gamma-Ray Burst 240219A and Its Implications. *ApJ*, volume 975, page L27, November 2024.

2024 Wan-Jin Lu, Ping Zhou, Pei Wang, **Shao, Yi-Xuan**, Xiang-dong Li, Jacco Vink, Di Li, and Yang Chen. Upper Limits on the Radio Pulses from Magnetars and a Central Compact Object with FAST. *ApJ*, volume 963, page 151, March 2024.

2024 Shi-Jie Gao, **Shao, Yi-Xuan**, Pei Wang, Ping Zhou, Xiang-Dong Li, Lei Zhang, Joseph W. Kania, Duncan R. Lorimer, and Di Li. Discovery of a Millisecond Pulsar Associated with Terzan 6. *ApJ*, volume 974, page L2, October 2024.

Presentations

2026.8 **Diverse Intensity–Polarization Relations in Pulsar Single Pulses**, 2026 Symposium on Frontiers and Technological Advances in Radio Astronomy, Delingha, China

2026.5 **Polarization-Dependent Distributions of Single Pulse Intensity Revealed by FAST**, APRIM2026, HongKong, China

2025.8 **The absence of radio emission reveals an exceptionally weak explosion for the probable historical supernova Pa 30**, The 16th Zhang Heng Academic Symposium, Qingdao, China

2024.12 **GTC Optical/Near-infrared Upper Limits and NICER X-Ray Analysis of SGR J1935+2154 for the Outburst in 2022**, PSR 2024, Guizhou, China

2024.12 **A pipeline to search for special magneto-active regions near Fast radio bursts**, FRB 2024, Khao Lak, Thailand

Research Experience

Nanjing University, *Advisor: Prof. Ping Zhou*

2023.9 – present ***Multiwavelength observation of isolated neutron stars***.

Utilizing NICER (X-ray), GTC/VLT (optical and near-infrared), and FAST (radio) to search for and study radiative emission from isolated neutron stars (focus on magnetars, central compact objects (CCOs), and high-magnetic-field pulsars). (Shao et.al, 2024, ongoing observations and 1 manuscript submitted...)

2024.7 – 2025.9 ***VLA observation of a remnant from double-white dwarf merger, Pa30***.

Analyzing the VLA imaging data, search for the extended radio emission, study the true nature of Pa30, SNR or a wind nebula? (Shao et.al, 2025)

Tsinghua University, *Advisor: Prof. Dongzi Li*

2024.7 – present ***FRB local environment and FAST single pulse polarization study***.

Study the neighbor plasma environment of some FRB source through polarization observation. Study the single pulse polarization characteristics, which may reveal the magnetosphere configuration. (1 manuscript submitted and ongoing observations...)

Awards & Scholarships

2024 Xiaomi Scholarship - CNY 5000

2023 Outstanding Graduate, Nanjing University

2022 Xiangyun Scholarship - CNY 7000

2022 Outstanding Award, Fundamental Sciences Scholarship, Nanjing University - CNY 5000

2021 Yang Lanyun Leadership Excellence Scholarship - CNY 5000

2020 Zheng Gang Elite Scholarship - CNY 2000

2020 Scholarship, National Astronomical Observatories of China - CNY 3000

Awarded Telescope time (**as PI**) till 2026.10.1

FAST 27.25h; 2022(6h), 2023(1h), 2024(2h), 2025(6h), 2026(8.75+3.5h)

GTC	1.5h; 2022(1.5h)
VLT	1.5h; 2025(1.5h)
XMM-Newton	55ks; 22ks for AO25+35ks for ToO in 2026
Parkes	39h; 2025(12h), 2026(27h)

Computer skills

Programming Languages	Python, Bash (daily use); C/C++ and Fortran (occasional use)

Teaching Assistantship

Fall, 2024 : **General Astronomy**.

Advisors & Collaborators

Ping Zhou
Associate Professor
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